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FOOD BANK INVENTORY MANAGEMENT SYSTEM

. Introduction

The Food Bank Inventory Management System is a console-based application developed using C++. The main purpose of this system is to manage food items stored in a food bank efficiently. It helps users to add, view, search, update, and delete food item records easily.

This system reduces manual work and improves inventory management by maintaining food details such as Food ID, Food Name, Quantity, and Expiry Date in an organized manner.

2. Objectives

- To maintain food inventory records efficiently.
- To reduce manual errors in food management.
- To provide quick searching of food items.
- To update food quantities easily.
- To remove expired or unnecessary food records.
- To learn Object-Oriented Programming concepts using C++.

3. System Features

Main Features

1. Add New Food Item
2. Display All Food Items
3. Search Food Item by ID
4. Update Quantity of Food Item
5. Delete Food Item
6. Menu Driven User Interface
7. Inventory Storage Using Arrays
8. Expiry Date Management
9. Easy Console-Based Operation

4. Technologies and Concepts Used

Technologies Used

- Programming Language: C++
- Platform: Console Application

- Compiler: GCC / g++
- IDE:
 - Visual Studio Code
 - Code::Blocks
 - Dev-C++

Concepts Used

- Classes and Objects
- Arrays of Objects
- Functions
- Loops (for, do-while)
- Conditional Statements (if, switch)
- String Handling
- Linear Search
- Input and Output Streams
- Object Assignment

5. System Design

Class Design

Class: Food Item

DATA MEMBERS	DESCRIPTION
ID	STORES FOOD ID
NAME	STORES FOOD NAME
QUANTITY	STORES QUANTITY
EXPIRY DATE	STORES EXPIRY DATE

Member Functions

FUNCTION	PURPOSE
ADD FOOD ()	ADDS FOOD DETAILS
DISPLAY FOOD ()	DISPLAY FOOD DETAILS

6. Workflow

Step-by-Step Working Process

1. Program starts.
2. Main menu is displayed.
3. User selects an option.
4. Based on choice:
 - Add food item
 - View items
 - Search item
 - Update quantity
 - Delete item
5. System performs selected operation.
6. Menu repeats until user chooses Exit.
7. Program terminates.

7. Sample Output

Main Menu

```
===== FOOD BANK INVENTORY MANAGEMENT =====
```

1. Add Food Item
2. View All Food Items
3. Search Food Item
4. Update Quantity
5. Delete Food Item
6. Exit

Add Food Item

```
Enter Food ID: 101
Enter Food Name: Rice Bags
Enter Quantity: 50
Enter Expiry Date: 20/12/2026
Food Item Added Successfully!
```

View Food Items

```
-----
Food ID      : 101
Food Name    : Rice Bags
Quantity     : 50
```

Expiry Date : 20/12/2026

Search Food Item

Enter Food ID to Search: 101

Food Item Found Successfully!

Update Quantity

Enter Food ID to Update Quantity: 101

Current Quantity: 50

Enter New Quantity: 70

Quantity Updated Successfully!

Delete Food Item

Enter Food ID to Delete: 101

Food Item Deleted Successfully!

Exit

Thank You for Using the System!